Ordinary Differential Equations: Abridged Notes

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These abridged notes were reconstructed afterward because I did not take direct notes during class. They are not a record of the actual lecture.

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1 Topic Summary

A repeated characteristic root supplies fewer distinct exponential solutions than the order of the equation requires. For a double real root r, the second independent solution is te^{rt}. Reduction of order is a more general method: when one nonzero solution y_1 of a second-order homogeneous linear equation is known, seek another as $y_2 = v(t)y_1(t)$. This method also works when coefficients vary with time.

2 Repeated Roots

If $ar^2 + br + c = a(r - r_0)^2$, then

$$\boxed{y(t) = (C_1 + C_2 t)e^{r_0 t}.}$$

The factor t makes $te^{r_0 t}$ independent of $e^{r_0 t}$. The two are a fundamental pair because their Wronskian is $e^{2r_0 t} \neq 0$.

2.1 Example: A Repeated-Root Initial-Value Problem

Adapted from Boyce et al., 11th ed., §3.4, Problem 9. Solve

$$9y'' - 12y' + 4y = 0, \qquad y(0) = 2, \qquad y'(0) = -1.$$

Solution.

The characteristic polynomial is $(3r-2)^2$, so the double root is $r_0 = 2/3$. Write $y = (C_1+C_2t)e^{2t/3}$. The first initial condition gives $C_1 = 2$. Since

$$y' = e^{2t/3}\left(C_2 + \frac{2}{3}(C_1 + C_2 t)\right),$$

the second gives $C_2 + 4/3 = -1$, hence $C_2 = -7/3$. Therefore

$$\boxed{y(t) = \left(2 - \frac{7}{3}t\right)e^{2t/3}.}$$

3 Reduction of Order

For a normalized equation $y'' + p(t)y' + q(t)y = 0$ and a known solution y_1 that is nonzero on the interval of work, substitute $y_2 = vy_1$. Simplifying with the equation satisfied by y_1 yields a first-order equation for v'. One convenient formula is

$$\boxed{y_2(t) = y_1(t)\int \frac{\exp\left(-\int p(t)\,dt\right)}{y_1(t)^2}\,dt.}$$

Any constant added inside the outer integral merely adds a multiple of y_1; choose the constant to obtain an independent solution.

3.1 Example: Variable Coefficients

Adapted from Boyce et al., 11th ed., §3.4, Problem 21. Given $y_1(x) = \sin(x^2)$, find a second solution of

$$xy'' - y' + 4x^3 y = 0, \qquad x > 0.$$

Solution.

After division by x, $p(x) = -1/x$, so $e^{-\int p\,dx} = x$ up to a constant factor. On an interval where $\sin(x^2) \neq 0$, reduction of order gives

$$\begin{aligned} y_2 &= \sin(x^2) \int \frac{x}{\sin^2(x^2)}\,dx \\ &= \frac{1}{2}\sin(x^2) \int \csc^2 u\,du, \qquad u = x^2 \\ &= -\frac{1}{2}\cos(x^2). \end{aligned}$$

A nonzero constant multiple is equally valid, so choose

$$\boxed{y_2(x) = \cos(x^2).}$$

This expression solves the equation for every $x > 0$, including zeros of $\sin(x^2)$; the intermediate integral is used locally where its denominator is nonzero. The pair is independent because its Wronskian is $-2x \neq 0$ for $x > 0$.